INCH-POUND
MIL-PRF-49176B(CR)
19 MAY 1997
SUPERSEDING
MIL-R-49176A(CR)
18 January 1982

PERFORMANCE SPECIFICATION
REGULATOR, BIAS VOLTAGE, CN-1503/UA

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Bias Voltage Regulator, CN-1503/UA, herein referred to as the regulator.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	- Standard General Requirements For Electronic Equipment

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4d, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Drawings listed below are to be used to ensure accurate interface of the item produced under this performance specification with existing hardware.

DRAWINGS

USA CECOM

SM-D-773912	- Schematic Diagram, Bias Regulator Assembly
SM-C-773541	- Connector, Receptacle, Electrical
SM-C-773501	- Connector, Receptacle, Electrical

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC S-815A	- General Requirements For Soldering Electronic Interconnections.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z 1.4	- Sampling Procedures And Tables For Inspection By

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, Wisconsin 53202.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.b), a sample(s) shall be subjected to first article inspection in accordance with paragraph 4.2.

3.2 Materials. The contractor shall select materials which shall be fully capable of meeting all the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The regulator provides low-noise regulated voltage used for biasing an array of detectors. The regulator is normally interconnected with an infrared system.

3.4 Design. The regulator shall meet the form, fit, function and interface requirements as specified herein.

3.4.1 Weight. The weight of the regulator shall be 0.16 (+0.02 -0.03) pound as weighed by a scale having an accuracy of 1% or better.

3.4.2 Dimensions The regulator printed wiring board shall be of appropriate material not thicker than 0.059 inches and shall have dimensions of 1.90 inches x 5.55 inches, $\pm$ 0.02 inches. When mounted, components shall not extend more than a maximum of 0.454 inches from the surface of the printed wiring board. When oriented horizontally, the two top corners of the board shall be rounded at 0.25 inches radius. The two bottom corners of the board shall be trimmed at a 45° angle between two points 0.25 inches vertical and horizontal of the corners, removing a triangle of board material 0.25 by 0.25 on either side of the right angle (corner). The x-axis datum line shall be located parallel to the lower edge of the board and 0.250 inches above that edge. It shall pass through the two 0.062 inch diameter holes located 0.250 in from the trimmed corners. The y-axis datum line shall pass through the left-hand 0.062 inch diameter hole and perpendicular to the x-axis datum line. Two mounting holes 0.140 inches in diameter shall be located 1.450 inches above the x-axis datum line and 0.050 inches to the left of the y-axis datum line and 5.100 inches to the right of the y-axis datum line. Two connectors shall be attached to the component side of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

board for the purpose of interfacing with existing hardware. One connector shall have the dimensions of Part Number SM-C-773541-3 as shown on Drawing SM-C-773541, while the other connector shall have the dimensions of Part Number SM-C-773501-3 in accordance with Drawing SM-C-773501. Two tip jacks shall also be attached to the component side of the board. These tip jacks shall mount in the two pairs of 0.052 inch diameter holes located 1.100 and 1.500 inches above the x-axis datum line and 1.100 and 3.900 inches to the right of the y-axis datum line. The tip jacks shall be JOLO part numbers LPCJ-419-03 and LPCJ-419-04 or M39024/11-03 and M39024/11-04 or equivalent are recommended to support interface with existing hardware. The pinout assignments for the connectors shall be in accordance with Drawing SM-D-773912 to ensure proper interface with existing hardware.

3.4.3 Components. The regulator shall contain manufactured components selected by the contractor as required to meet the performance specifications contained herein.

3.5 Performance characteristics.

3.5.1 Output noise. Output noise shall be not greater than 23×10^{-9} vrms/$\sqrt{\text{Hz}}$ when measured per 4.5.1.

3.5.2 Output voltage and output current. The unloaded regulator output voltage shall be +5.0, ±0.2 volts direct current (Vdc) at 23°, ±5°C and shall be +5.0, ±0.5 Vdc over the temperature range of -54°C to +71°C, with an input of +9.5 to +10.0 Vdc. The output voltage, loaded with 4.64 ohms, ±1 percent resistance shall not vary from the unloaded output voltage by more than ±50 millivolts dc over the temperature range. Output voltage and current shall be measured per 4.5.2 to verify that the regulator is capable of supplying detectors at 6 milliamperes each, or a total of 1.08 amperes minimum.

3.5.3 Current limit. Short-circuit current limit shall be 1.60, (+0.30 -0.35) amps, when measured with a short circuit on the regulator output.

3.5.4 Line rejection. Alternating current (ac) ripple rejection at the output, referenced to the input, shall not be less than 50 dB from 10 Hz to 35 KHz, with the regulator loaded (see 3.5.1). Line rejection may be measured by modulating the input supply with a sinusoidal signal, which provides a variable input ripple. Line rejection is the ratio of the input peak-to-peak ripple to output peak-to-peak ripple. Line rejection shall be measured from 10 KHz to 35 KHz at the following frequencies: 10 Hz, 100 Hz, 1 KHz, 10 KHz, and 35 KHz.

3.5.5 Output impedance. Output impedance shall be less than 0.3 ohms up to 50 KHz. The output impedance is the ratio of the output voltage difference between the no load and full load conditions to the maximum output current at full load condition.

3.6 Environmental conditions.

3.6.1 Temperature shock. The regulator shall not be damaged (see 6.3.1) by sudden changes in temperature between -54°C and +95°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

3.6.2 High temperature. The regulator shall not be damaged by storage to +95 °C or continuous operation to +71 °C, and shall operate for 30 minutes at +95 °C.

3.6.3 Low temperature. The regulator shall not be damaged by operation to -54 °C or storage to -62 °C.

3.6.4 Shock. The regulator shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling inspections.

3.6.5 Vibration. The regulator shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in Figure 1.

3.7 Reliability. The regulator shall have a mean time between failure (MTBF) of at least 39,100 hours, θ_o.

3.8 Coating. Unless otherwise specified, the regulator and its components and parts shall be cleaned and coated to prevent corrosion, oxidation, and fungus growth.

3.9 Preconditioning. Each regulator shall be subjected to at least a 20-cycle preconditioning environment as specified in Figure 2 prior to Group A inspection (for exception to this requirement see 6.2.g).

3.10 Soldering. Unless otherwise specified, soldering shall conform to ANSI/IPC S-185A.

3.11 Nameplates, and product marking. The regulator, parts, components, subassemblies and assemblies thereof shall be marked for identification and reference designation markings. For guidance, use MIL-HDBK-454.

3.12 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

4.2 First article testing. When specified in the contract, the first article inspection shall be performed by the contractor as specified in the contract. First article reliability (4.2.3) is not required unless specified in the contract (see 6.2.b.3).

4.2.1 Inspections. All materials, parts, processes and assemblies shall be visually examined for conformance to the applicable specification or drawing and Table I inspections. Inspection shall be made using sampling per ASQC Z 1.4.

TABLE I. Visual inspection.

Inspection	Requirement paragraph	Test paragraph
Any part or component missing or damaged	3.4	4.2.1
Dimensions not as specified Length Width	3.4	4.2.1
Weight not as specified	3.4.1	4.2.1
Coating not as specified	3.8	4.2.1
Soldering not as specified	3.10	4.2.1
Marking not as specified	3.11	4.2.1
Workmanship not as specified	3.12	4.2.1

* In ASQC Z 1.4 use inspection level S-3 for objective quality evidence.

4.2.2 Tests. Upon successful completion of inspections specified in 4.2.1, perform the inspections of Table II in any order.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Output noise	3.5.1	4.5.1
Output voltage and output current	3.5.2	4.5.2
Current limit	3.5.3	4.5.3
Line rejection	3.5.4	4.5.4
Output impedance	3.5.5	4.5.5
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Reliability	3.7	4.7

4.2.3 First article reliability. The lower inspection MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

4.2.3.1 Inspection level.

- Temperature: +50°C, (+5 -0)°C
- Temperature cycle: none
- Vibration: 2.2 g's, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be perpendicular to the plane of the printed circuit board.
- Equipment on-off cycling: turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout this requirement verification.

4.2.3.2 Parameters. The discrimination ratio θ_0/θ_1 shall be 3.

4.2.3.3 Failure criteria and test length. The inspections of Table III shall be performed at least once prior to and once after the inspection of 4.2.3.1. Additional inspections of Table III may be selected by the contractor. The results of all Table III inspections and any physical damage shall be used to determine the success or failure of the reliability requirement. In the event of a Table III inspection failure, the time interval between the preceding successful inspections and the failure shall not contribute to the total cumulative hours. Verification shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with Figure 5. Only equipment "ON" time shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half of the average operating time of all equipment "ON" inspections.

4.2.4 Disposition of first article samples First article samples shall not be considered as part of the procurement quantities (see 6.2.c).

4.3 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the regulator shall be operated at +23°C, ±5°C. The input voltage shall be +9.5 to +10.0 Vdc. Unless otherwise specified, all inspection set-up and load resistor tolerances shall be ±5 percent.

4.4 Conformance inspection.

4.4.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of +23°C, ±5°C on all regulators following the preconditioning of 3.9 and the inspection of 4.2.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Output noise	3.5.1	4.5.1
Output voltage and output current	3.5.2	4.5.2
Current limits	3.5.3	4.5.3

4.4.2 Group B inspection. Not required.

4.4.3 Group C inspection. Group C inspections shall be conducted on regulators selected from units which have passed the tests in 4.2.1. The sample(s) shall be tested in accordance with the inspections specified in Table IV. Samples shall be selected in accordance with the contract (see 6.2.c).

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Line rejection	3.5.4	4.5.4
Output impedance	3.5.5	4.5.5
Vibration	3.6.5	4.6.5

4.5 Methods of inspection.

4.5.1 Output noise. Output noise voltage shall be determined by measuring wideband rms noise voltage at the regulator output with a true-rms voltmeter using the test setup and test methods of 4.5.2. Since the noise output of the regulator is very low, it may fall under the minimum threshold level of the true-rms voltmeter. The measurement may then be performed using a low-noise amplifier connected at the +5 Vdc output of the regulator. Using the voltage gain and signal bandwidth of the amplifier, noise shall be calculated using the equation as follows:

$$e_{noise} / \sqrt{Hz} = [e_{out} \div (A_V \times (NBW)^{1/2})]$$

where: $e_{noise} / \sqrt{Hz}$ = equivalent input noise seen at the regulator output

e_{out} = rms noise voltage measured at the amplifier output

A_V = amplifier voltage gain, typically 6000 V/V; equivalent input noise typically 2nV / $\sqrt{Hz}$

NBW = noise bandwidth = $\pi/2$ x signal bandwidth

signal bandwidth low 3dB point =10 Hz maximum

signal bandwidth high 3dB point = 35 kHz minimum

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

4.5.2 Output voltage and output current. Output voltage and current shall be measured to verify that the regulator is capable of supplying 180 detectors at 6 mA each, or a total of 1.08 amperes. A resistive load of 4.64 ohms, ± 1 percent shall simulate the detector load, and both the loaded and unloaded output voltages shall be measured. The loaded output voltage shall be measured for input supply voltages of +9.5 and +10.0 Vdc. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Current limit. Short-circuit current limit shall be measured with an ammeter with a short circuit on the regulator output. Failure to meet requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Line rejection. With the regulator loaded (4.6.1), line rejection shall be measured by modulating the input supply with a sinusoidal signal, which provides a variable input ripple. An oscilloscope shall be used to observe the ripple inserted on the input and the ripple appearing on the output. The oscilloscope shall be isolated and referenced to test point J10 on the regulator. Line rejection is the ratio of the input peak-to-peak ripple to output peak-to-peak ripple. Line rejection shall be measured from 10 Hz to 35 KHz. Failure to meet requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Output impedance. Output impedance shall be measured using a varying load network with a modulated input and an oscilloscope to measure the ac output current and voltage. The output impedance is the ratio of the output voltage difference between the no load and full load conditions to the maximum output current at full load condition. Failure to meet requirements of 3.5.5 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing methods contained in section 6.4 are acceptable methods for determining compliance with Section 3.6 et el. The contractor may elect to use alternative methods and testing procedures. In such cases, the contractor shall be required to show where his own tests and measurements for verifying the environmental conditions are equivalent to the methods cited herein. Where the contractor desires to deviate from these cited methods, he shall be prepared to show equivalency via a test plan which has been approved by the Government contracting office. Also, environmental tests shall be accomplished using known commercially accepted testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device, or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.

4.6.1 Temperature shock. Measure compliance with the requirement of 3.6.1. Recommend performing the tests prescribed in 6.4. Failure to meet the requirement shall constitute failure of the sample.

4.6.2 High temperature. Measure compliance with the requirement of 3.6.2. Recommend performing the tests prescribed in 6.4. Failure to meet the requirement shall constitute failure of the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

4.6.3 Low temperature. Measure compliance with the requirement 3.6.3. Recommend performing the tests prescribed in 6.4. Failure to meet the requirement shall constitute failure of the sample.

4.6.4 Shock. Measure compliance with the requirement of 3.6.4. Recommend performing the tests prescribed in 6.4. Failure to meet the requirement shall constitute failure of the sample.

4.6.5 Vibration. Measure compliance with the requirement of 3.6.5. Recommend performing the tests prescribed in 6.4. Failure to meet the requirement shall constitute failure of the sample.

4.7 Reliability.

4.7.1 First article reliability. The lower test MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs:

4.7.1.1 Inspection level.

- Temperature: +50°C, (+5 -0)°C
- Temperature cycle: None
- Vibration: 2.2 g's, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be perpendicular to the plane of the printed circuit board.
- Equipment on-off cycling: Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout this requirement verification.

4.7.1.2 Parameters. The discrimination ratio θ_0/θ_1 shall be 3.

4.7.1.3 Failure criteria and inspection length. The inspections of Table III shall be performed at least once prior to and once after the inspection of 4.2.3.1. Additional inspection of Table III may be selected by the contractor. The results of all Table III inspections and any physical damage shall be used to determine the success or failure of the reliability requirement. In the event of a Table III inspection failure, the time interval between the preceding successful inspections and the failure shall not contribute to the total cumulative hours. Verification shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with Figure 5. Only equipment "on" time

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half the average operating time of all equipment "on" inspections.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.e). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The regulator is intended for use in infrared systems.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required:

(1) Time frame for submission of first article test reports.

(2) Time frame for approval of the first article test reports.

(3) First article reliability testing, if required.

c. Disposition of first article samples.

d. Sampling plans for Group C inspection (see 4.4.3).

e. Necessary actions by the contractor in the event of a Group C failure (see 4.4.3.1).

f. Packaging requirements (see 5.1).

g. The preconditioning requirement of 3.9 is required for procurement of all modules to this specification made directly for the Government. Implementation of 3.9 may be waived by the systems contractors ordering modules to this specification for use in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

manufacture of complete systems or subassemblies thereof if these systems are subjected to equivalent preconditioning.

h. Environmental pollution prevention measures are contained in the packaging material specifications or provided by preparing activity for recommended disposability.

i. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, and 2.3).

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of regular performance to values less than specified herein.

6.4 Environmental inspection procedures. The test methods and procedures of MIL-STD-810, listed below, are recommended, but are not mandatory for determining compliance with the environmental requirements of this specification.

TABLE V. Environmental inspection procedures.

Inspection	Req. Para.	MIL-STD-810 Method	MIL-STD-810 Procedure	MIL-STD-810 Remarks
Temperature shock	3.6.1	503	I	Non-operating mode. In step 2 use -54°C, and for step 1 and 4 use +95°C
High temperature (see Figure 3)	3.6.2	501	I	Test simultaneously with paragraph 3.5.2, for final 30 minutes of step 3. For step 2 use +95°C and Steps 4 & 5 use +71°C
Low temperature (see Figure 4)	3.6.3	502	I	Test simultaneously with paragraph of 3.5.2, in lieu of Table III inspection. For step 2 use -62°C and step 4 use -54°C.
Shock	3.6.4	516.1	IV, V	Non-operating mode.
Vibration (see Figure 1)	3.6.5	514.1	I, VIII	Non-operating mode. Use Figure 1 (page 11) in lieu of MIL-STD-810 curve. Sinusoidal cycle time shall be 120 minutes per axis. Dwell time shall be 1/6 of the cycling time at each resonance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing.

Infrared System
Detector Array

6.6 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

(Project 5855-0111)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

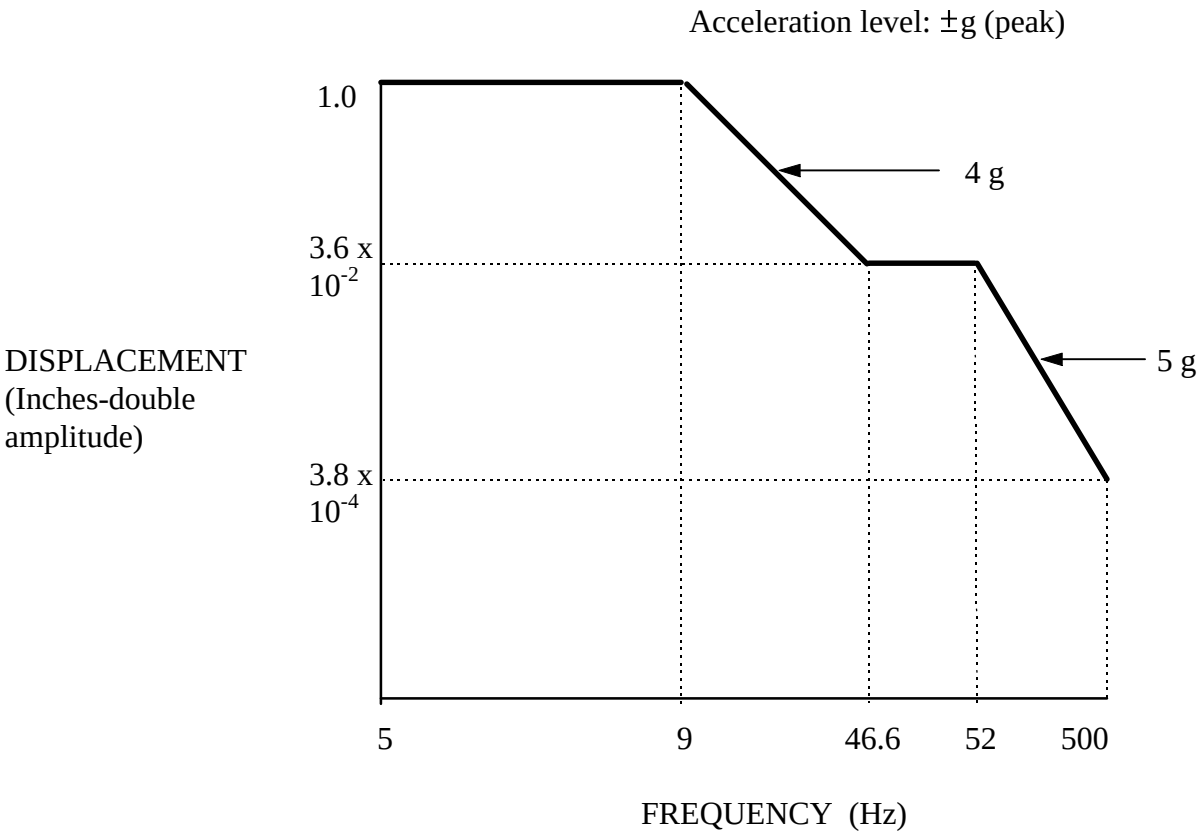


FIGURE 1. Vibration test profile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

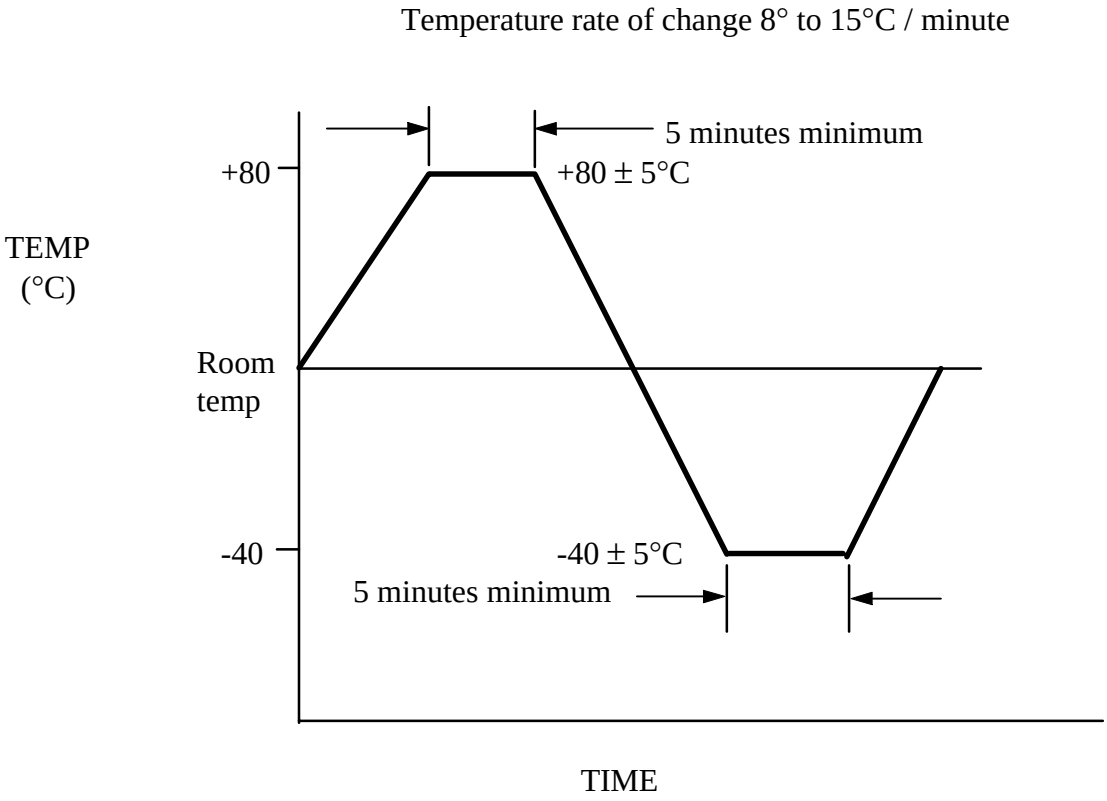
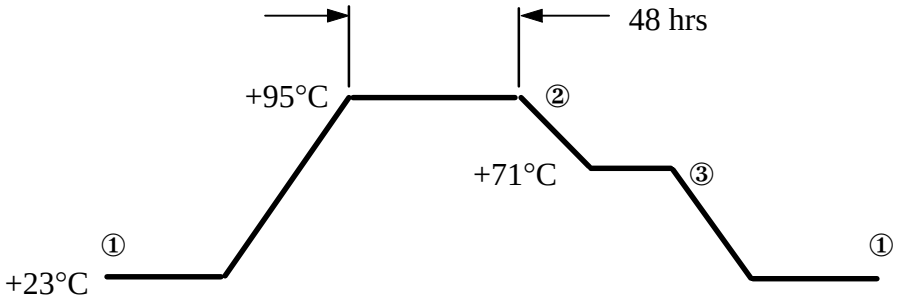


FIGURE 2. One preconditioning cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.



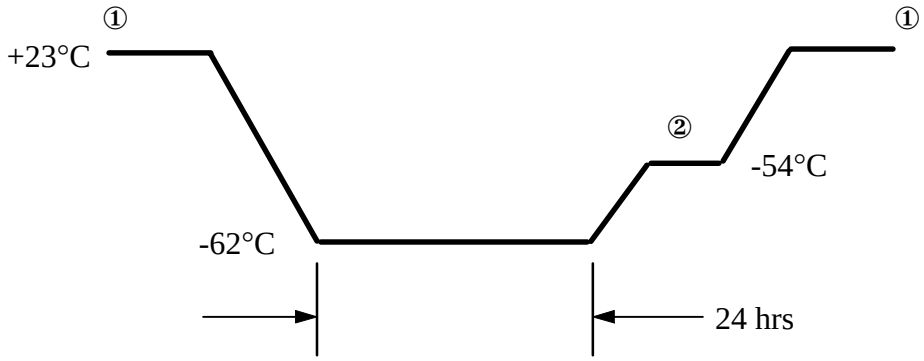
① Group A tests (Table III) are connected at these points.

② Test item is operated for 30 minutes at this point.

③ Output voltage and output current tests (3.5.1) are conducted at this point.

FIGURE 3. High temperature test profile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

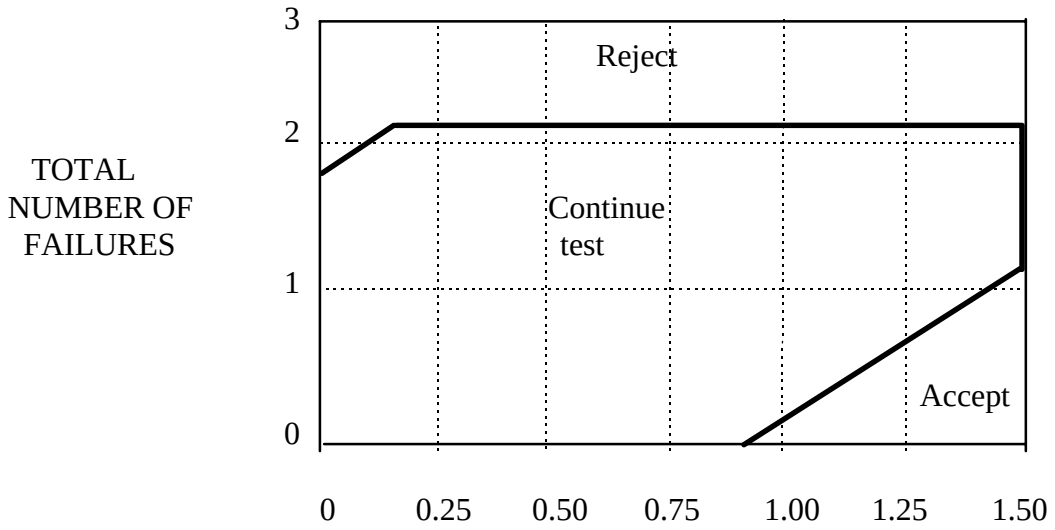


① Group A tests (Table III) are conducted at these points

② Output voltage and output current tests (3.5.1) are conducted at this point.

FIGURE 4. Low temperature test profile.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.



TOTAL TEST TIME*

Number of failures	Reject (equal or less)	Accept (equal or more)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*Total test time is the total unit hours of equipment "ON time" and is expressed in multiples of lower test MTBF, θ_1

FIGURE 5. First article reliability accept - reject criteria.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49176A	**2. DOCUMENT DATE (YYMMDD)** 970519

3. DOCUMENT TITLE REGULATOR, BIAS VOLTAGE, CN-1503/UA

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE	b. TELEPHONE *Include Area Code)* (1) Commercial (732) 532-2735 (2) AUTOVON 992-2735
c. ADDRESS *(Include Zip Code)* US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:54Z
Check the source to verify that this is the current version before use.